Group project 6

Md. Zahidul Islam, Jiajie Wu, Meiheng Ren

May 15st, 2022

1 Description of Approach

We put different multiplier values in the last assignment and did not use any iterative process to update the multiplier values. We know that the Lagrange function for the problem is as follows:

$$\text{Let } L(\lambda) = min \sum_{i,t,q} P_{iq} X_{itq} - \sum_{t} \lambda_t (K_t - \sum_{i} w_{it}),$$

To find the optimal lambda values, we need to solve another maximization problem, that is

$$\max_{\lambda \geq 0} L(\lambda)$$

As the $L(\lambda)$ is a piecewise concave function we can use the subgradient method to find the λ which maximizes the value of $L(\lambda)$. Here, λ is the vector of multipliers.

To solve the above problem iteratively we need to follow the following iterative process

$$\lambda_{j+1,t} = max(0, \lambda_{j,t} + \alpha_{j,t} \bigtriangledown L(\lambda) = \lambda_{j,t} + \alpha_{j,t} \times (k_t - \sum_{i} w_{it}))$$

,

where i = item number, t = time periods, j = iteration number.

$\alpha_{j,t}$ is an important value to change iteration-wise. We use $\alpha_{j,t} = \frac{1}{j \times sensitivity}$. After trying to use many sensitivity values, we found sensitivity = 1000000. During updating the λ, we track the objective values for the Lagrange problem as the lower bound of the problem and if we put the solution of the Lagrange problem to the actual objective function, we will get the upper bound of the problem. We track the problem and saw that with increasing the iteration number, the gap between the lower bound and the upper bound closer and closer. We also calculate the gap and track the progress. We set the total number of iterations to 200. All the lambda values will not create a feasible solution. We will track all those lambda values that produces a feasible solution for the actual problem.

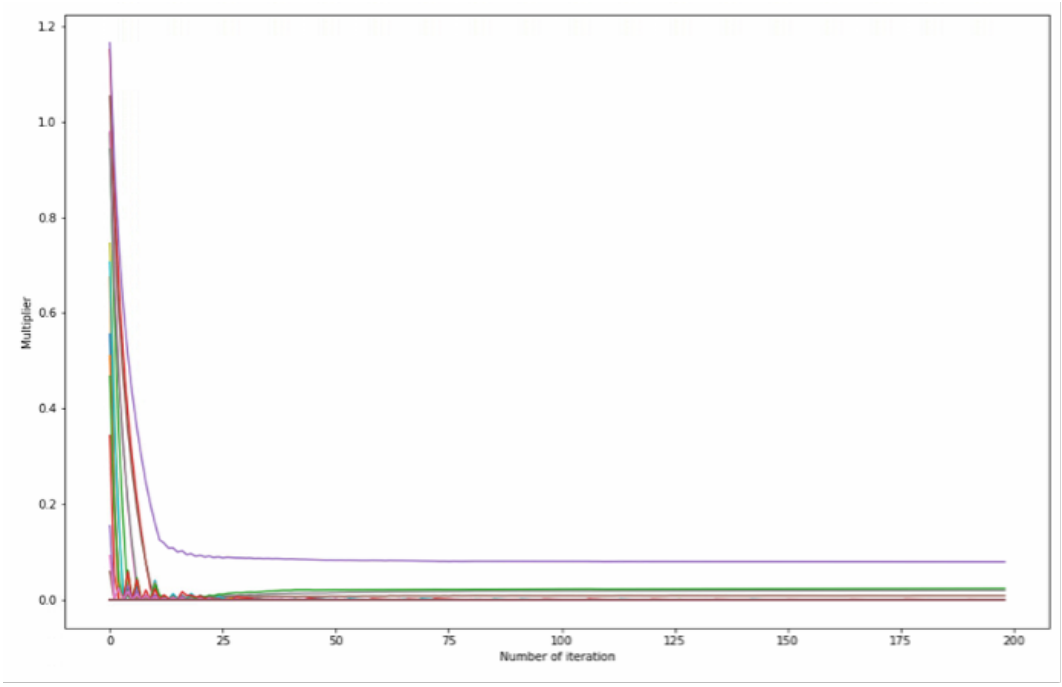
2 Discussion of Result

2.1 The objective and corresponded multiplier

The actual objective is 2485690.64, and the objective of the Lagrange problem is 2485508.7. The actual objective is slightly higher than the relaxed objective, which keeps the consistency of the theory. The final %gap between the upper and lower bounds is 0.06% for the problem after 200 run and 550 seconds runtime. The lowest gap was 1% in 1200 seconds when we solved the problem with the actual formulations. So, the Lagrangian relaxation is working well.The corresponding multiplier vector is shown below:

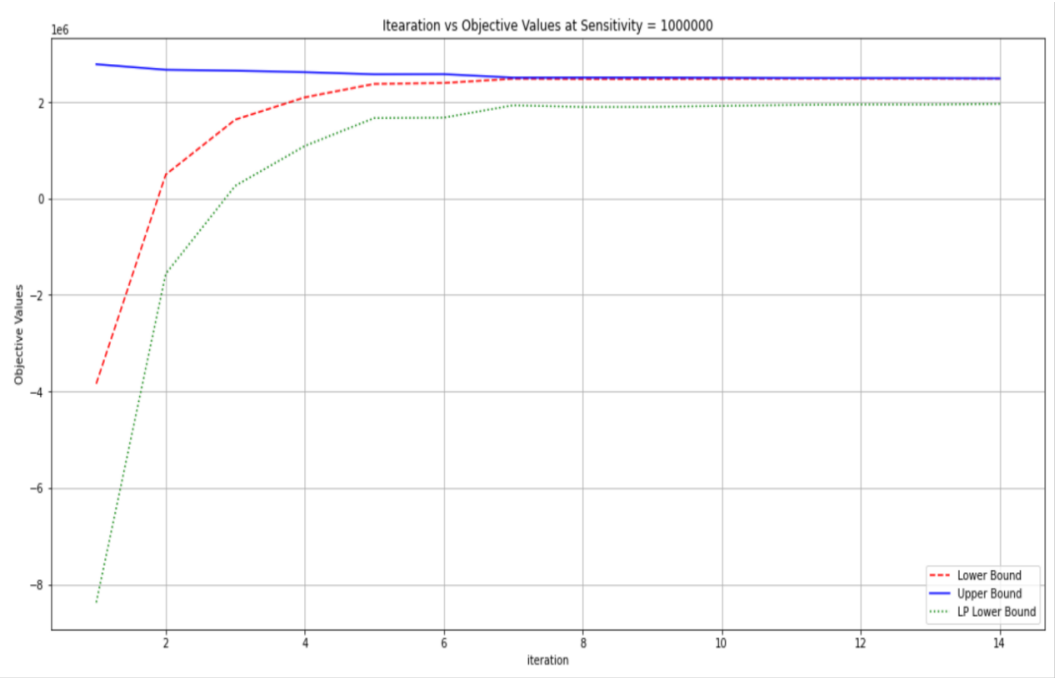
The relaxed item	Multiplier
1	0
2	0
3	0
4	0.0001
5	0.0794
6	0.0089
7	0.0202
8	0.0202
9	2.4926
10	0.0002
11	0
12	0
13	0.0240
14	0.0002
15	0.0003
16	0.0002
17	0.004
18	0
19	0
20	0
21	0
22	0
23	0
24	0
25	0
26	0

2.2 The update of multiplier

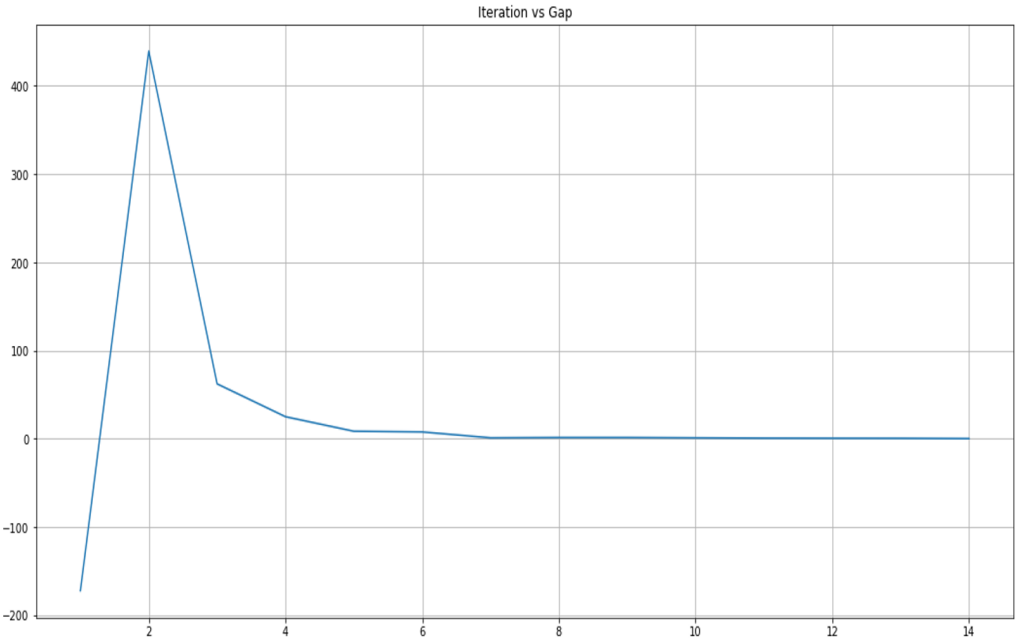


As we can see from the figure, the multiplier value drops drastically initially, and then each multiplier converges to a low level.

2.3 The comparison between each bound



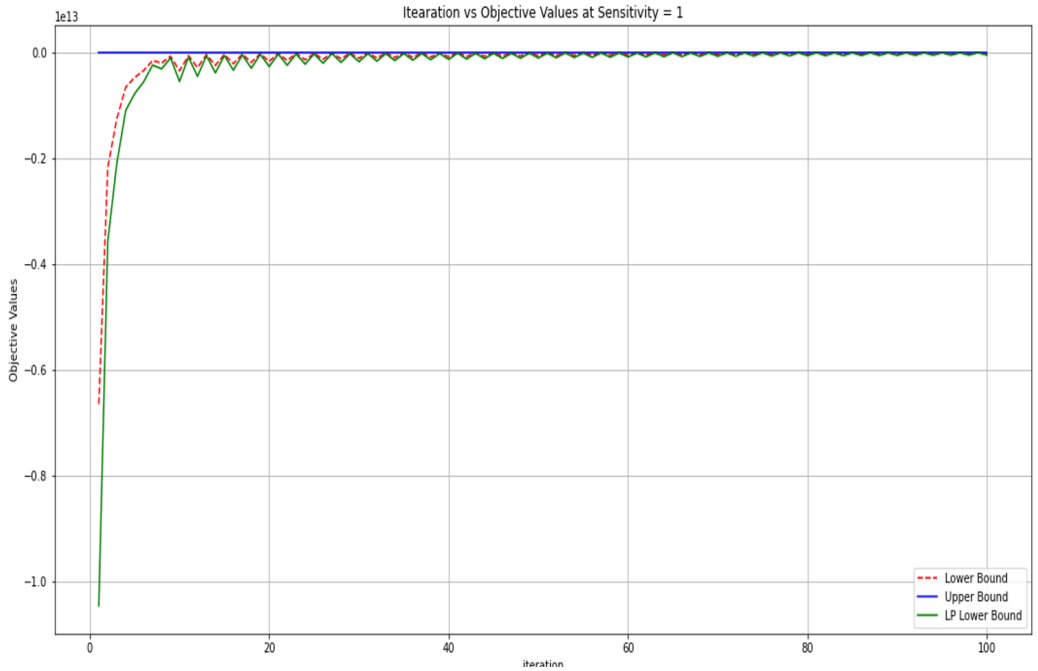
As we can see from the figure, the blue,red and green line refers to the upper bound, lower bound and LP lower bound respectively. It is very intuitive and clear to see that the lower bound of LP-relaxation is lower than the lower bound.

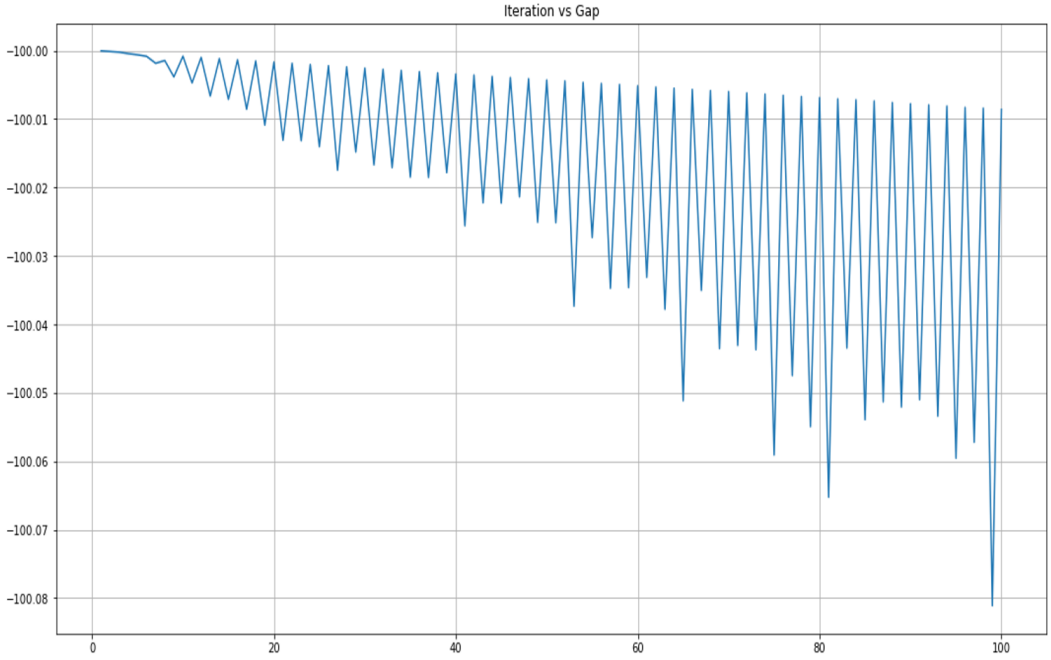


From the above figure, we can see that the %gap ((Upper Bound- Lower Bound)×100/Lower Bound) reduces with increase in iteration and goes to close to zero eventually.

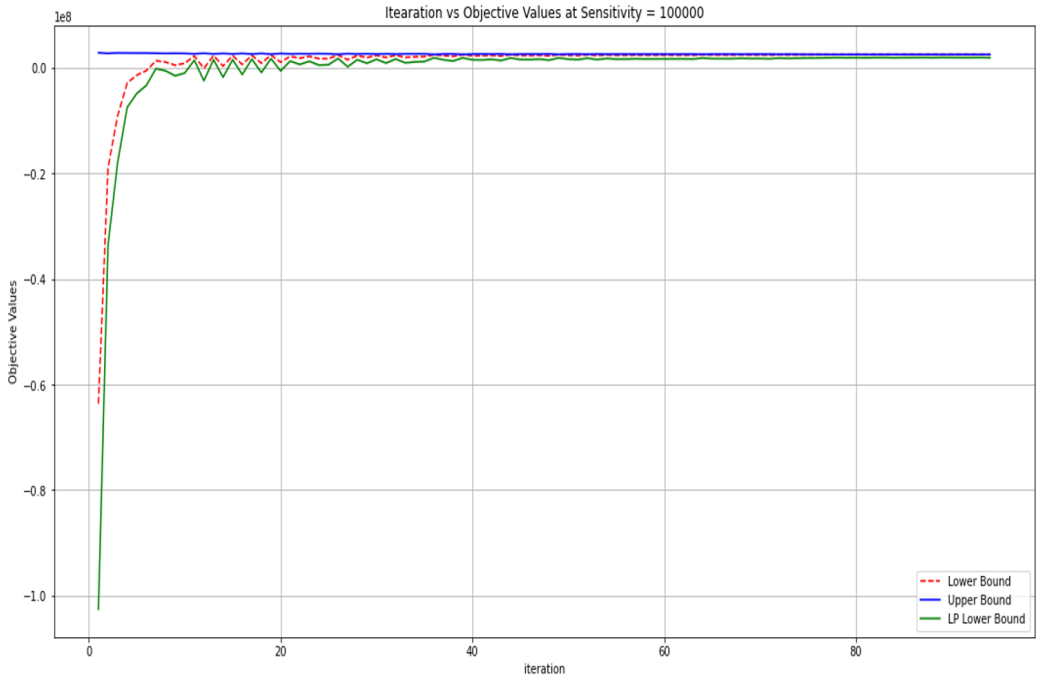
Choosing *sensitivity* in $\alpha_{j,t} = \frac{1}{j \times sensitivity}$ is an important work in the iterative.

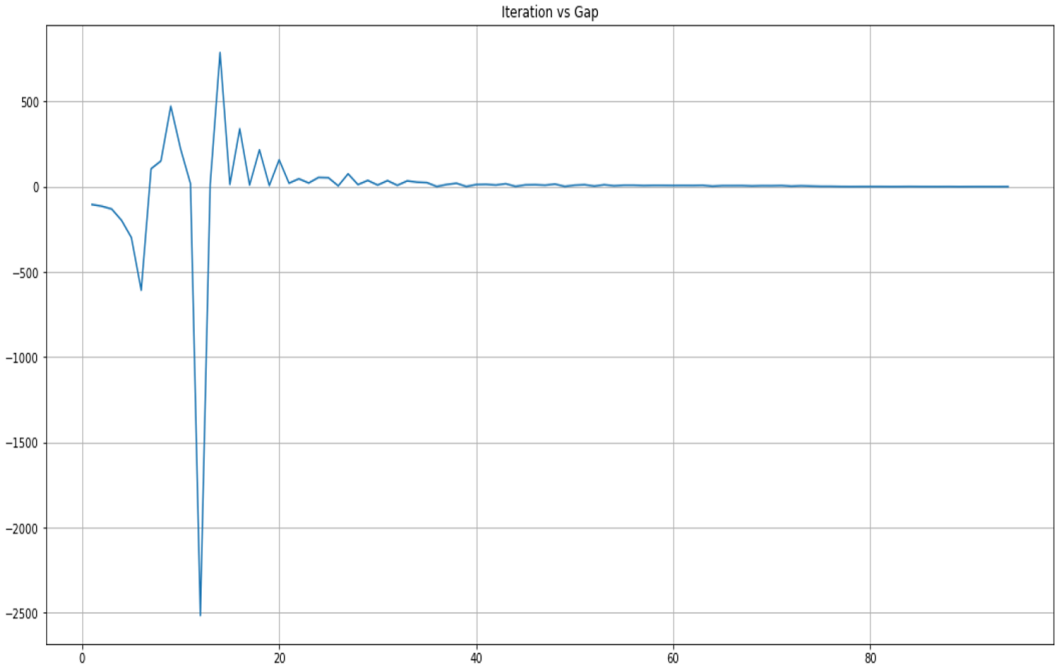
We can see that when the $sensitivity = 1$, the problem does not converge



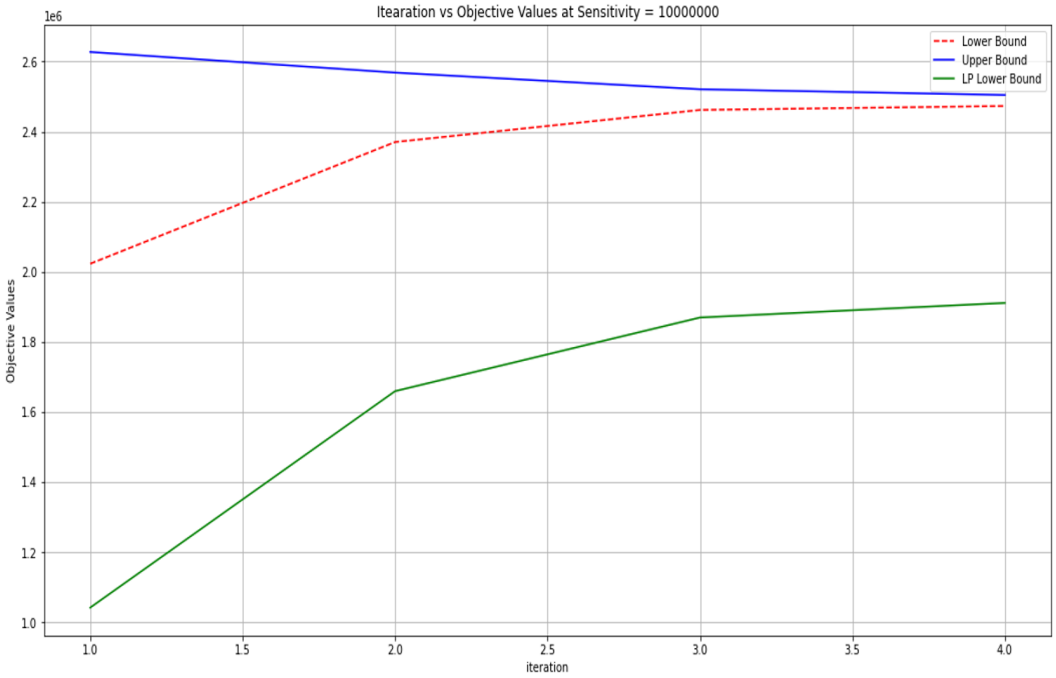


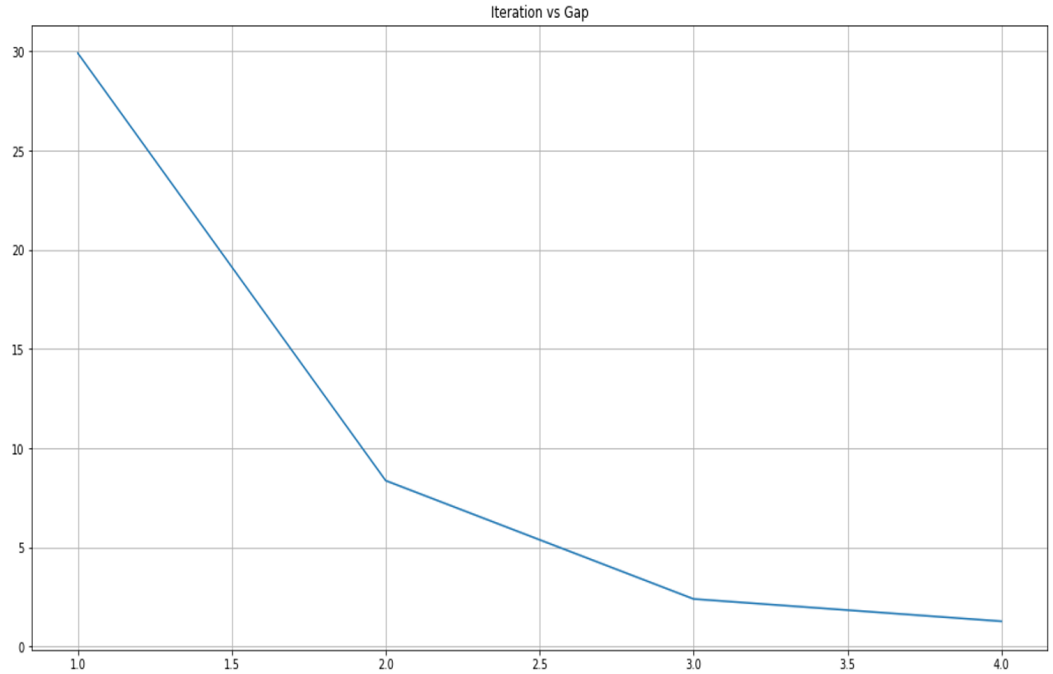
The problem will not converge until the value of sensitivity is 100000.



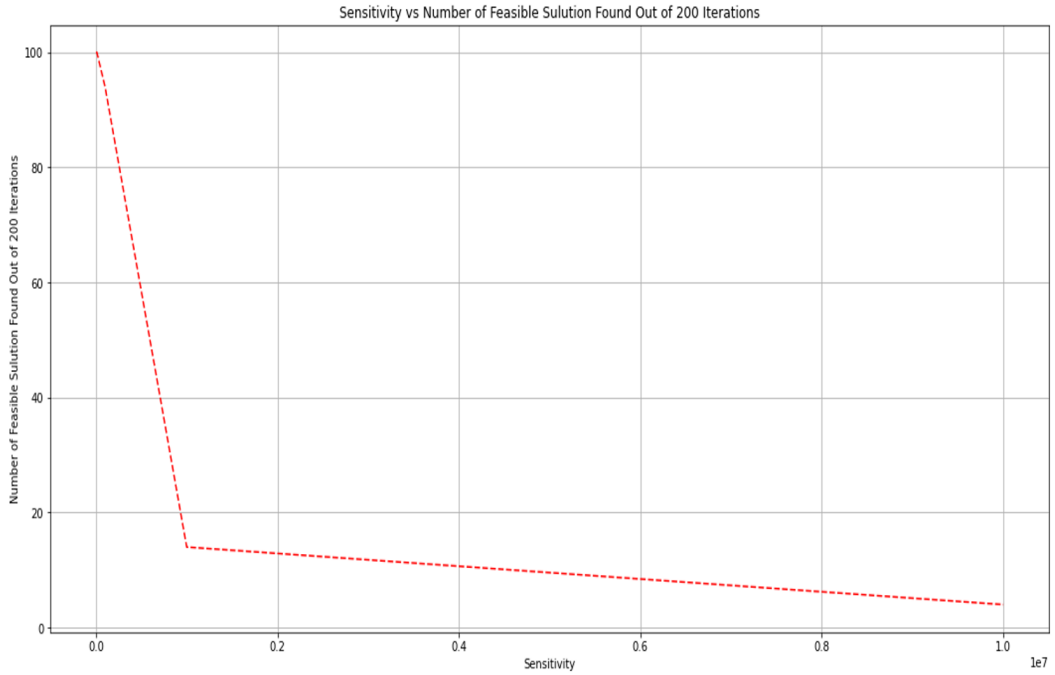


For higher sensitivity value (10000000), it will take a greatly high amount of time to converge as shown below.





Another observation is that with an increase in sensitivity value the number of feasible solutions found by the algorithm also decreases as shown in the figure below:



The runtime increases if we use higher lambda values (decreasing the learning rate for the gradient descent method).

